

BOWLING GREEN, KY, USAWMO: **746716**Lat: **36.965N**Lon: **86.424W**Elev: **528**StdP: **14.42**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **93808****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.6	16.8	0.6	5.8	16.2	5.4	7.4	19.5	23.5	56.9	20.5	52.1	4.8	330	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	94.1	75.2	91.7	75.1	89.9	74.6	78.5	88.9	77.5	87.7	76.5	86.0	7.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	136.0	83.9	74.5	131.5	82.8	73.4	126.9	81.7	42.3	89.2	41.2	87.7	40.2	86.1	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	17.5	15.6	5.8	98.4	5.7	4.7	1.7	101.8	-1.7	104.6	-4.9	107.2	-9.1	110.7		
(5)			4.6	80.7	5.4	1.6	0.7	81.8	-2.4	82.8	-5.5	83.7	-9.4	84.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.7	36.3	40.7	48.6	58.6	67.4	75.4	78.8	77.7	71.4	59.7	48.1	40.2
	DBStd	17.17	11.93	10.81	10.32	8.98	7.42	5.23	4.44	4.62	6.71	8.55	9.15	10.22
	HDD50	1422	451	297	152	26	1	0	0	0	0	17	141	337
	HDD65	3892	891	681	515	229	64	3	0	0	21	208	510	770
	CDD50	4580	25	36	109	284	541	761	891	859	641	317	84	32
	CDD65	1580	0	1	7	37	140	314	426	395	212	44	3	1
	CDH74	15571	0	4	61	342	1275	3061	4405	3900	2030	467	26	0
	CDH80	6164	0	0	5	54	365	1222	1914	1680	807	114	3	0
Wind	WSAvg	6.0	7.2	7.3	7.4	7.5	6.2	5.4	4.9	4.4	4.3	5.0	5.7	6.5
Precipitation	PrecAvg	50.0	3.7	4.1	4.9	4.2	5.0	4.1	4.3	3.6	3.7	3.0	4.3	4.7
	PrecMax	76.6	10.4	11.3	14.6	9.7	13.7	11.1	8.9	8.3	11.7	7.7	8.4	13.1
	PrecMin	31.3	0.9	0.7	1.7	0.9	1.4	1.1	1.2	0.7	0.3	0.1	0.7	1.1
	PrecStd	8.6	2.1	2.2	2.6	2.1	2.7	2.4	2.1	1.9	2.4	1.8	1.9	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.6	73.1	80.0	84.6	89.4	93.8	97.9	98.2	95.8	87.3	77.5	70.2
		MCWB	61.6	61.7	64.4	67.3	72.8	74.8	77.8	75.4	71.8	70.5	65.2	62.5
	2%	DB	64.3	67.9	75.0	81.3	86.5	91.5	93.4	94.0	91.7	83.1	72.3	65.1
		MCWB	58.9	58.9	61.4	65.7	71.0	74.8	77.0	75.1	72.4	67.7	61.3	59.0
	5%	DB	60.7	63.7	70.6	78.2	84.2	89.6	91.2	91.0	88.4	79.4	68.3	61.1
		MCWB	55.7	55.7	58.9	64.2	69.7	74.2	76.4	74.9	72.0	65.6	59.1	55.7
	10%	DB	55.8	59.0	66.3	74.7	81.6	87.2	89.1	88.6	84.6	75.3	64.0	56.9
		MCWB	50.5	52.5	56.1	62.3	69.3	73.3	75.5	74.3	70.4	64.5	56.3	51.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.3	64.3	66.4	70.9	75.7	78.7	80.5	79.4	76.7	74.3	67.3	64.2
		MCDB	66.5	70.5	75.2	80.8	84.8	88.8	91.4	91.6	87.4	82.4	73.3	68.2
	2%	WB	60.2	61.1	63.7	68.6	73.6	77.3	79.1	77.9	75.2	71.2	64.2	60.1
		MCDB	63.7	66.6	73.0	77.8	83.2	87.8	89.5	88.6	85.6	79.0	70.1	64.2
	5%	WB	56.7	57.4	60.7	66.4	71.9	76.1	78.0	76.9	73.9	68.6	60.9	56.4
		MCDB	60.3	62.1	68.6	74.6	81.3	85.9	88.2	86.7	83.4	75.9	66.5	60.3
	10%	WB	51.0	52.9	57.8	64.2	70.4	74.9	77.0	75.8	72.5	66.1	57.4	52.5
		MCDB	55.1	58.0	64.9	72.3	79.0	84.0	86.6	85.0	81.4	73.4	62.5	56.5
Mean Daily Temperature Range	5% DB	MDBR	17.0	18.4	20.5	21.9	20.2	19.3	18.5	19.8	21.8	22.7	20.5	16.7
		MCDBR	20.3	24.4	26.2	25.1	23.0	22.1	20.9	22.5	25.8	26.9	26.1	21.1
		MCWBR	15.7	17.3	15.9	13.4	10.4	8.6	7.6	8.1	9.5	13.2	16.7	16.5
	5% WB	MCDBR	18.8	20.8	22.3	20.5	19.6	19.0	18.3	18.9	20.2	21.2	20.7	17.8
	MCWBR	16.1	16.9	14.9	12.6	10.1	8.5	7.8	7.8	8.9	12.0	15.5	15.8	
Clear-Sky Solar Irradiance	taub	0.317	0.325	0.369	0.407	0.450	0.472	0.506	0.517	0.427	0.369	0.333	0.310	
	taud	2.462	2.423	2.346	2.243	2.168	2.141	2.065	2.018	2.256	2.404	2.464	2.489	
	Ebn at Noon	277	288	284	277	266	259	249	243	261	268	268	272	
	Edh at Noon	27	31	37	43	47	49	52	54	40	31	26	24	
All-Sky Solar Radiation	RadAvg	661	857	1203	1592	1780	1980	1889	1751	1467	1119	789	569	
	RadStd	64	112	113	111	160	118	112	91	125	128	82	56	